The heavy-quarkonium hyperfine splittings as the arbiter of the contact interaction

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With the ATLAS measurement of the B_c^* mass, every S-wave hyperfine splitting of the $c\bar{c}$, $b\bar{b}$ and $c\bar{b}$ ground and first-excited states is now known, together with the P-wave singlet–triplet offsets of h_c, $h_b(1P)$ and $h_b(2P)$: nine observables that test the spin–spin interaction of a heavy quark and antiquark across three reduced masses and two radial excitations. We confront them with four prescriptions for the contact term—first-order perturbation theory, a mass-independent Gaussian smearing, the mass-dependent smearing of Godfrey and Isgur, and a Yukawa form factor with the mass scale of the lattice spin–spin potential—each with a fixed or a one-loop running coupling, on top of a spin-averaged potential and quark masses fixed beforehand by the spin-independent spectrum, so that nothing but the contact prescription is tested. The data are decisive on three points. The h_c, which sits at the χ_{cJ} centroid to 0.2 MeV, limits the r.m.s. radius of any smearing to below 0.07 fm, an order of magnitude smaller than the widths used in standard potential models and small enough that smearing has no effect the data can see. The $c\bar{c}/b\bar{b}$ ratio of the $1S$ splittings, together with the $2S/1S$ ratios, fixes the short-distance strength κ of the Coulomb term, to which the spin-averaged spectrum is blind: with a running coupling the splittings require $\kappa \simeq 0.2$–0.4, consistent with $\frac{4}{3}\alpha_s$, whereas with a fixed coupling they require $\kappa = 0$. And every prescription fails in the same place: the charmonium $2S$ splitting is predicted at 55–63 MeV against the measured 48.3, the one state among the nine for which coupled-channel shifts of this size have been computed. With that state removed, first-order perturbation theory with a running coupling and no smearing describes the other eight with $\chi^2 = 2.8$ for seven degrees of freedom and a single parameter, $\Lambda = 0.21$ GeV, and gives $M(B_c^*) - M(B_c) = 62 \pm 1 \pm 3$ MeV and $M(B_c^*(2S)) - M(B_c(2S)) = 33 \pm 1 \pm 2$ MeV (64.5 and 34.4 with the annihilation-corrected charmonium input): in agreement with ATLAS and CMS, and 2σ above the lattice value of 54(3) MeV.

1 Introduction

The spin–spin interaction between a heavy quark and antiquark is, at leading order, a contact term: one-gluon exchange gives

$$H_{SS} = \frac{32\pi\alpha_s}{9\, m_1 m_2}\, \mathbf{s}_1 \cdot \mathbf{s}_2\, \delta^3(\mathbf{r}), \tag{1}$$

so that the S-wave hyperfine splitting is proportional to $|\psi(0)|^2/m_1 m_2$ and the P-wave singlet sits at the triplet centroid. In first-order perturbation theory this works tolerably for charmonium and bottomonium separately but has always been in tension between them [1, 2], and because a δ function is singular when inserted into a Schrödinger equation, most potential models replace it by a smeared version, Gaussian with a width of order 1 GeV [2, 3], and solve the singlet and triplet non-perturbatively. The width is then a parameter, mass-independent in some models and mass-dependent in others, and its value has been set by the charmonium $1S$ splitting alone.

Until recently the data could not arbitrate. What has changed is that the B_c^* has been found [4], giving a third reduced mass; the $B_c(2S)$ doublet has been resolved [5, 6]; the $\eta_b(2S)$ exists; and the h_c and h_b masses are known to 0.1–1 MeV. Nine hyperfine observables are now measured (Table 1), spanning reduced masses from 0.9 to 2.6 GeV and two radial excitations, and they test the mass dependence, the radial dependence and the range of the contact interaction independently. This paper asks what they say.

The question is only sharp if nothing else is being fitted. We therefore take the spin-averaged potential $V(r)$ and the quark masses from a model-independent extraction from the spin-averaged $c\bar{c}+b\bar{b}$ spectrum [7], hold them fixed, and fit only the one to three parameters of each contact prescription to the nine splittings. Section 2 sets this up, Sec. 3 gives the comparison, Sec. 4 the consequences for B_c, and Sec. 5 concludes.

Table 1: The hyperfine data (MeV) [4, 5, 8]. S waves: $M(^3S_1) - M(^1S_0)$. P waves: $M(^1P_1)$ minus the $(2J+1)$-weighted centroid of the ^3P_J triplet, which vanishes for a pure contact term. Theory floors of 3 MeV (S) and 0.5 MeV (P) are added in quadrature in the fits.

observable	value	source
$c\bar{c}\,1S$	112.81 ± 0.30	$J/\psi - \eta_c$
$c\bar{c}\,2S$	48.30 ± 0.60	$\psi(2S) - \eta_c(2S)$
$b\bar{b}\,1S$	61.70 ± 2.00	$\Upsilon(1S) - \eta_b$
$b\bar{b}\,2S$	24.40 ± 3.50	$\Upsilon(2S) - \eta_b(2S)$ (Belle)
$B_c\,1S$	64.50 ± 1.70	$B_c^* - B_c$, ATLAS
$B_c\,2S$	35.40 ± 2.40	$B_c^*(2S) - B_c(2S)$, ATLAS+CMS
h_c	0.07 ± 0.15	$h_c - \bar{M}(\chi_{cJ})$
$h_b(1P)$	-0.57 ± 0.80	$h_b(1P) - \bar{M}(\chi_{bJ}(1P))$
$h_b(2P)$	-0.44 ± 1.20	$h_b(2P) - \bar{M}(\chi_{bJ}(2P))$

2 Set-up

Spin-averaged potential. Reference [7] represents $V(r) = -\kappa/r + S(r)$, S a cubic spline, and fits it with m_c and m_b to the eleven spin-averaged $c\bar{c}$ and $b\bar{b}$ multiplet centroids. The spectrum determines $S(r)$ to 40–70 MeV for $0.7 \le r \le 7\,\mathrm{GeV}^{-1}$ but is blind to the Coulomb coefficient: fits at $\kappa = 0$, 0.2, 0.4, 0.6 and

0.8 have $\chi^2 = 12.0$–12.2 for the same eleven centroids, with $m_c = 1.78$–1.81 and $m_b = 5.16$–$5.18\,\mathrm{GeV}$. These five potentials, each with its own masses, are the fixed input here, and κ is the one property of $V(r)$ that the hyperfine data, which probe $|\psi(0)|^2$, are in a position to decide.

Prescriptions. All four have the structure of Eq. (1) with $\delta^3(\mathbf{r})$ replaced by a normalised profile $\delta_X(r)$:

- *first-order*: $\delta_X = \delta^3$, the splitting evaluated in first-order perturbation theory, $\Delta = 8\alpha_s|R(0)|^2/9m_1m_2$; no P-wave shift. No free parameter beyond the coupling.
- *Gaussian*: $\delta_X = (\sigma/\sqrt{\pi})^3 e^{-\sigma^2 r^2}$, one width for all systems, as in Ref. [3]. One parameter, σ.
- *Godfrey–Isgur*: a Gaussian whose width depends on the quark masses, $\sigma_{ij}^2 = \sigma_0^2\big[\frac{1}{2} + \frac{1}{2}\big(\frac{4m_im_j}{(m_i+m_j)^2}\big)^4\big] + s^2\big(\frac{2m_im_j}{m_i+m_j}\big)^2$ [2]. Two parameters, σ_0 and s.
- *Yukawa*: $\delta_X = m^2 e^{-mr}/4\pi r$, the form that describes the lattice spin–spin potential of Koma and Koma [9] with m the 0^{-+} glueball mass, 2.47 GeV. One parameter, m, with $m = 2.47\,\mathrm{GeV}$ as a check.

For the three smeared forms the singlet ($\mathbf{s}_1\cdot\mathbf{s}_2 = -\frac{3}{4}$) and triplet ($+\frac{1}{4}$) radial equations are solved non-perturbatively, for S and P waves alike, with the Numerov method of Ref. [10]; the P-wave observable is then the singlet energy minus the triplet one, the spin–orbit and tensor terms cancelling in the centroid. The coupling is either one value α_s for all three systems ("fixed") or the one-loop running coupling at $\mu = \sqrt{m_1m_2}$ with Λ fitted ("running"), which makes $\alpha_s(b\bar{b})/\alpha_s(c\bar{c}) \simeq 0.68$. Each prescription is fitted at each κ; fits with the $c\bar{c}$ $2S$ splitting removed, and with theory floors of 1 and 5 MeV, are the variants. Propagated errors are scaled by $\sqrt{\chi^2/\mathrm{dof}}$ where that exceeds one.

3 What the data decide

Tables 2 and 3 and Fig. 1 summarise the fits. No prescription describes all nine observables: the best χ^2 is 15–22 for six to eight degrees of freedom, and the pattern of pulls is the same for all of them. Three features of the data account for this, and each settles a question.

The h_c excludes any visible smearing. A smeared contact term shifts P-wave singlets below the triplet centroid by $\langle\delta_X\rangle_P$, which grows with the width: the h_c is predicted 16 MeV below the χ_{cJ} centroid for $\sigma = 1\,\mathrm{GeV}$ and 4 MeV below for $\sigma = 2\,\mathrm{GeV}$, with α_s in each case tuned to the $c\bar{c}$ $1S$ datum (Table 4, Fig. 2). The measured offset is $+0.07 \pm 0.15\,\mathrm{MeV}$. Smearing widths of $\sigma \simeq 1$–$1.2\,\mathrm{GeV}$, which are what Refs. [3] and the charmonium literature use and which predict h_c masses of 3516–3517 MeV, are excluded by two orders of magnitude; the data allow a r.m.s. radius below about 0.07 fm ($\sigma \gtrsim 3.5\,\mathrm{GeV}$), the $\sigma = 3\,\mathrm{GeV}$ offset of $-1.6\,\mathrm{MeV}$ being 3σ away even with the 0.5 MeV theory floor. Every fit in Table 2 finds this on its own: the Gaussian and Yukawa widths run to $\sigma = 4.3$–$4.8\,\mathrm{GeV}$ and $m = 7$–$8\,\mathrm{GeV}$, radii of 0.05–0.07 fm, where the profile is narrower than any feature of the wave functions (Fig. 3). The lattice scale $m = 2.47\,\mathrm{GeV}$ is excluded for the same reason (h_c at $-5\,\mathrm{MeV}$). The widths do not run further for a reason worth stating: a δ function has no non-perturbative solution in three dimensions, the attractive singlet channel collapsing as the width shrinks, so the smeared prescriptions have no first-order limit. At $\sigma = 4\,\mathrm{GeV}$ the non-perturbative $c\bar{c}$ $1S$ splitting is already 25% above first order for the same α_s and the $b\bar{b}$ one 20% below it; at $\sigma = 8\,\mathrm{GeV}$ the $c\bar{c}$ ratio is 3 and at 16 GeV it is 260. The fitted widths sit at the edge of this regime, and what the fits with smearing are really doing is using the onset of the collapse as a free parameter. Smearing, in short, is not something the data can see; it is a regulator, and the hyperfine splittings tell us to remove it.

The $c\bar{c}/b\bar{b}$ ratio fixes the Coulomb strength, given the running. The ratio of the $b\bar{b}$ to the $c\bar{c}$ $1S$ splitting, 0.547 ± 0.018, is $|R_{b\bar{b}}(0)|^2 m_c^2 \alpha_s(b\bar{b})/|R_{c\bar{c}}(0)|^2 m_b^2 \alpha_s(c\bar{c})$ in first order, and $|R(0)|^2$ for the heavier system grows faster with κ. Table 5 and Fig. 4 show the χ^2 of each prescription against the five values of κ the spectrum allows. With a fixed coupling, first-order perturbation theory wants $\kappa = 0$—no Coulomb term at all—and is excluded at any $\kappa \geq 0.2$ ($\chi^2 \geq 55$); with the running coupling, whose $\alpha_s(b\bar{b})/\alpha_s(c\bar{c}) = 0.68$ supplies two-thirds of the mass dependence, it wants $\kappa = 0.2$–0.4 and is excluded at $\kappa \geq 0.6$. That range is $\frac{4}{3}\alpha_s$ for $\alpha_s = 0.15$–0.3, the one-gluon-exchange value, which the fixed-coupling fit cannot accommodate. The smeared prescriptions with a fixed coupling do the opposite, preferring $\kappa \geq 0.6$, because the non-perturbative response to the near-δ term, which enhances $c\bar{c}$ and reduces $b\bar{b}$, supplies mass dependence of the wrong sign that a stronger Coulomb term must undo; with a running coupling the mass-independent Gaussian and Yukawa forms fail at every κ ($\chi^2 \geq 80$), since both effects then act in the same direction on $b\bar{b}$, while the Godfrey–Isgur form survives by making its width mass-dependent ($s \simeq 2$), which is to say by putting the mass dependence back by hand. The short-distance strength of the potential and the running of the coupling are therefore not separately determined by the hyperfine data; they are determined together, and the physically motivated combination—$\kappa \simeq \frac{4}{3}\alpha_s$ with α_s running—is the one the data accept.

The $c\bar{c}$ $2S$ splitting is the universal outlier. Every fit in Table 3 predicts the $\psi(2S)$–$\eta_c(2S)$ splitting at 55–63 MeV against the measured 48.3 ± 0.6, and ev-

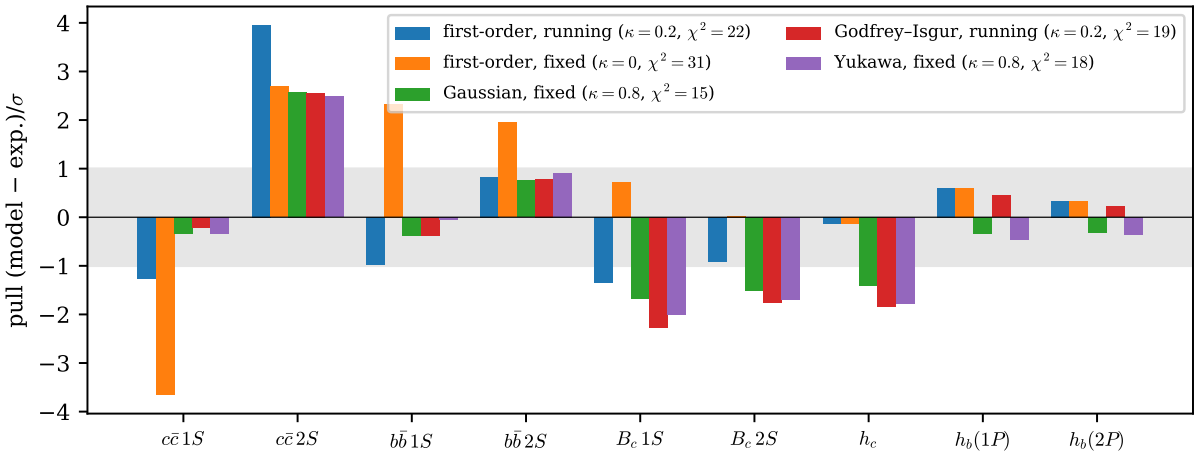


Figure 1: Pulls of the nine hyperfine observables for each prescription at the Coulomb strength κ that suits it best (Table 2). The grey band is $\pm1\sigma$. Every prescription over-predicts the $c\bar{c}$ $2S$ splitting.

Table 2: The four prescriptions with fixed and running coupling, each at the κ of Table 5 that minimises its χ^2: fit quality for the nine observables, fitted parameters (GeV), and the r.m.s. radius of the contact term for $c\bar{c}$ (and $b\bar{b}$ where it differs). Errors scaled by $\sqrt{\chi^2/\mathrm{dof}}$.

prescription	α_s	κ	χ^2/dof	parameters	r.m.s. radius [fm]
first-order	running	0.2	22.1/8	$\Lambda = 0.19 \pm 0.01$	–
first-order	fixed	0	30.9/8	$\alpha_s = 0.33 \pm 0.01$	–
Gaussian	fixed	0.8	14.8/7	$\alpha_s = 0.28 \pm 0.03$, $\sigma = 4.35 \pm 0.77$	0.056
Gaussian	running	0	102.9/7	$\Lambda = 0.12 \pm 0.05$, $\sigma = 4.83 \pm 1.40$	0.050
Godfrey–Isgur	running	0.2	19.2/6	$\Lambda = 0.15 \pm 0.05$, $\sigma_0 = 0.20 \pm 8.98$, $s = 2.16 \pm 0.63$	0.062 / 0.022
Godfrey–Isgur	fixed	0	17.4/6	$\alpha_s = 0.28 \pm 0.04$, $\sigma_0 = 3.81 \pm 0.88$, $s = 1.42 \pm 0.75$	0.053 / 0.029
Yukawa	fixed	0.8	17.5/7	$\alpha_s = 0.27 \pm 0.03$, $m = 6.97 \pm 1.32$	0.069
Yukawa	running	0	116.0/7	$\Lambda = 0.12 \pm 0.04$, $m = 7.59 \pm 1.88$	0.064

ery $c\bar{c}$ $2S/1S$ ratio comes out 0.50–0.55 against 0.43. This is not a failure the contact prescription can fix: the ratio is set by $|R_{2S}(0)|^2/|R_{1S}(0)|^2$, which the spin-averaged potential determines, and by second-order effects that are small for any profile the h_c allows. The $b\bar{b}$ ratio, 0.40 ± 0.06, and the B_c ratio, 0.55 ± 0.04, are both reproduced (predicted 0.46–0.49 and 0.49–0.53), and the lattice $b\bar{b}$ ratio, 0.445 ± 0.028 [11], agrees with the fits better than the experimental one does. The $c\bar{c}$ $2S$ states are the only ones among the nine within 50 MeV of an open-flavour threshold, and Eichten, Lane and Quigg [12] computed the coupled-channel shifts for exactly these two: +15.7 MeV for the $\eta_c(2S)$ and −5.2 MeV for the $\psi(2S)$, reducing their splitting by 20.9 MeV. Our 12–15 MeV excess is that effect. We therefore treat the $c\bar{c}$ $2S$ splitting as a coupled-channel observable rather than a contact-term one, and refit without it.

One parameter describes the rest. With the $c\bar{c}$ $2S$ removed, first-order perturbation theory with the running coupling fits the remaining eight observables with $\chi^2 = 2.8$ for seven degrees of freedom at $\kappa = 0.2$ (9.0 at $\kappa = 0.4$), with one parameter, $\Lambda = 0.213 \pm 0.011$ GeV, i.e. $\alpha_s(c\bar{c}) = 0.35$, $\alpha_s(b\bar{b}) = 0.24$, $\alpha_s(c\bar{b}) = 0.29$ (Table 6). The smeared forms do no better ($\chi^2 = 7$–11 for five or six degrees of freedom) with more parameters. One more correction is known and is not in any potential model: the η_c annihilation shift. The lattice value of the $c\bar{c}$ $1S$ splitting for the connected contribution alone is 120.3 ± 1.1 MeV [13], the difference from experiment being the $\eta_c \to gg$ shift of 7.3 ± 1.2 MeV. A contact term compared with experiment is therefore being asked to reproduce 7 MeV of annihilation physics, and the right comparison for it is the connected value. Fitting to 120.1 ± 1.2 instead of 112.8 gives $\chi^2 = 2.1/7$ with $\Lambda = 0.236 \pm 0.011$ GeV (last row of Table 6); the $b\bar{b}$ splittings are unchanged and the B_c ones move up by 2–3 MeV.

4 The B_c hyperfine splittings

The B_c is the system where the prescriptions differ most, because it is the only one not used to set the smearing width's mass dependence, and it is where experiment and lattice QCD disagree: ATLAS finds $M(B_c^*) - M(B_c) = 64.5 \pm 1.7$ MeV [4], HPQCD predicted 54±3 [14], a 3σ difference. Table 6 gives the predictions of the acceptable fits. They span 57–63 MeV: the first-order running-coupling fits, which are the only ones with $\chi^2/\mathrm{dof} \simeq 1$, give 62.1 ± 1.2 ($\kappa = 0.2$) and

Table 3: Predicted splittings (MeV) for the fits of Table 2, in the order of that table.

	$c\bar{c}\,1S$	$c\bar{c}\,2S$	$b\bar{b}\,1S$	$b\bar{b}\,2S$	$B_c\,1S$	$B_c\,2S$	h_c	$h_b(1P)$	$h_b(2P)$
experiment	112.8 ± 0.3	48.3 ± 0.6	61.7 ± 2.0	24.4 ± 3.5	64.5 ± 1.7	35.4 ± 2.4	0.1 ± 0.1	-0.6 ± 0.8	-0.4 ± 1.2
first-order, running	109.0	60.4	58.2	28.2	59.9	31.9	0.0	0.0	0.0
first-order, fixed	101.8	56.5	70.1	33.4	67.0	35.5	0.0	0.0	0.0
Gaussian, fixed	111.8	56.2	60.3	27.9	58.7	29.6	-0.7	-0.9	-0.9
Gaussian, running	115.5	58.5	36.0	15.8	47.0	23.5	-0.6	-0.6	-0.6
Godfrey–Isgur, running	112.1	56.1	60.3	28.0	56.7	28.6	-0.9	-0.1	-0.1
Godfrey–Isgur, fixed	110.3	55.9	65.7	29.6	58.1	28.9	-0.6	-0.3	-0.3
Yukawa, fixed	111.8	55.9	61.5	28.6	57.6	28.9	-0.9	-1.0	-0.9
Yukawa, running	115.5	58.1	34.5	15.0	45.6	22.6	-0.8	-0.7	-0.7

Table 4: Gaussian contact term of width σ with α_s tuned to reproduce the $c\bar{c}$ $1S$ splitting ($\kappa = 0.4$): the predicted $c\bar{c}$ $2S$ splitting and h_c offset (MeV). Measured: 48.3 and $+0.07 \pm 0.15$.

σ [GeV]	r.m.s. [fm]	α_s	$c\bar{c}\,2S$	$M(h_c) - \bar{M}(\chi_{cJ})$
0.5	0.483	2.78	33.6	−46.33
0.7	0.345	1.50	33.6	−29.29
1	0.242	0.89	39.6	−16.05
1.5	0.161	0.58	47.2	−7.28
2	0.121	0.45	51.3	−3.97
3	0.081	0.35	55.0	−1.62
4	0.060	0.30	56.4	−0.84
5	0.048	0.26	56.4	−0.49

Table 5: χ^2 of each prescription (nine observables) against the Coulomb strength κ of the spin-averaged potential. The spin-averaged spectrum has $\chi^2 = 12.0$–12.2 at every κ.

prescription	α_s	$\kappa = 0$	0.2	0.4	0.6	0.8
first-order	running	–	22.1	23.6	44.1	90.7
first-order	fixed	–	54.6	94.8	155.1	238.0
Gaussian	fixed	–	22.7	18.9	16.2	14.8
Gaussian	running	–	–	82.2	–	–
Godfrey–Isgur	running	–	19.2	21.7	28.9	35.9
Godfrey–Isgur	fixed	–	–	23.5	–	–
Yukawa	fixed	–	28.4	23.0	19.1	17.5
Yukawa	running	–	–	91.1	–	–

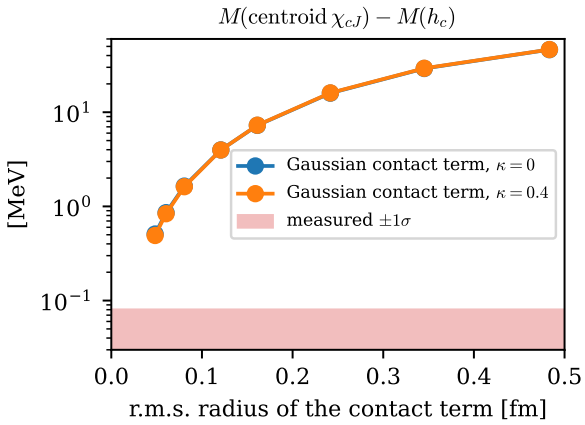


Figure 2: The h_c offset below the χ_{cJ} centroid produced by a Gaussian contact term, as a function of its r.m.s. radius, with α_s tuned to the $c\bar{c}$ $1S$ splitting; the band is the measurement.

63.1 ± 1.4 ($\kappa = 0.4$); the smeared forms, whose second-order singlet enhancement is largest for the most compact system, $b\bar{b}$, and is therefore tuned away for $c\bar{c}$ and $b\bar{b}$ but not for $c\bar{b}$, give 57–60. Taking the best fit with the measured inputs as central and the spread of the first five rows as the systematic,

$$M(B_c^*) - M(B_c) = 62 \pm 1 \pm 3\,\text{MeV}, \quad (2)$$

$$M(B_c^*(2S)) - M(B_c(2S)) = 33 \pm 1 \pm 2\,\text{MeV}, \quad (3)$$

$$\Delta_{1S} - \Delta_{2S} = 29.0 \pm 0.5 \pm 0.7\,\text{MeV}, \quad (4)$$

against the measured 64.5 ± 1.7, 35.4 ± 2.4 and 29.1 ± 1.7. The prediction agrees with ATLAS at 0.7σ and lies 2.2σ above the lattice value, and with the annihilation-corrected $c\bar{c}$ input it becomes 64.5 ± 1.2 and 34.4 ± 0.6 MeV, on top of the measurements. The hyperfine data of the equal-mass systems, interpolated with the prescription they themselves select, side with ATLAS. The B_c P-wave singlet–triplet offset is predicted to vanish to 0.5 MeV, which will be a clean test of the same point once the $B_c(1P)$ multiplet is seen.

5 Conclusions

The nine measured heavy-quarkonium hyperfine observables, confronted with four contact-interaction prescriptions on top of a spin-averaged potential fixed by the spectrum, decide the following:

1. The h_c offset from the χ_{cJ} centroid, 0.07 ± 0.15 MeV, limits the r.m.s. radius of any smeared contact term to below 0.07 fm. The $\sigma \simeq 1$ GeV Gaussians of the potential-model literature are excluded by two orders of magnitude, and the fits themselves push every smearing to the point where it is invisible.

2. With a one-loop running coupling, the $c\bar{c}/b\bar{b}$ hyperfine ratio selects a Coulomb strength $\kappa = 0.2$–0.4 in the spin-averaged potential, consistent with $\frac{4}{3}\alpha_s$ and invisible to the spectrum itself; with a

Table 6: Fits without the $c\bar{c}$ $2S$ splitting, and the resulting B_c predictions (MeV). Errors are the propagated fit errors scaled by $\sqrt{\chi^2/\mathrm{dof}}$; the spread of the central values across the five rows is the prescription systematic. Experiment: $\Delta_{1S} = 64.5 \pm 1.7$ [4], $\Delta_{2S} = 35.4 \pm 2.4$ [4, 5], $\Delta_{1S} - \Delta_{2S} = 29.1 \pm 1.7$ (CMS [5]), 30.9 (LHCb [6]); lattice $\Delta_{1S} = 54 \pm 3$ [14]. The last two columns are the predicted $c\bar{c}$ $2S$ splitting and B_c P-wave offset.

fit	χ^2/dof	Δ_{1S}	Δ_{2S}	$\Delta_{1S} - \Delta_{2S}$	$M(B_c^*)$	$c\bar{c}\,2S$	$B_c\,1P$
first-order, running, $\kappa = 0.2$	2.8/7	62.1 ± 1.2	33.1 ± 0.6	29.0 ± 0.5	6336.6	63.2	+0.0
first-order, running, $\kappa = 0.4$	9.0/7	63.1 ± 1.4	33.8 ± 0.7	29.3 ± 0.6	6337.6	61.2	+0.0
Gaussian, fixed, $\kappa = 0.8$	6.9/6	59.6 ± 1.7	30.2 ± 0.6	29.4 ± 1.3	6334.1	57.6	-0.5
Yukawa, fixed, $\kappa = 0.8$	10.1/6	58.2 ± 2.3	29.4 ± 0.7	28.8 ± 1.7	6332.7	57.4	-0.6
Godfrey–Isgur, running, $\kappa = 0.2$	11.1/5	57.1 ± 3.2	29.0 ± 1.5	28.1 ± 1.8	6331.6	58.0	-0.2
first-order, running, $\kappa = 0.2$, lattice $c\bar{c}\,1S$	2.1/7	64.5 ± 1.2	34.4 ± 0.6	30.2 ± 0.6	6339.0	66.3	+0.0

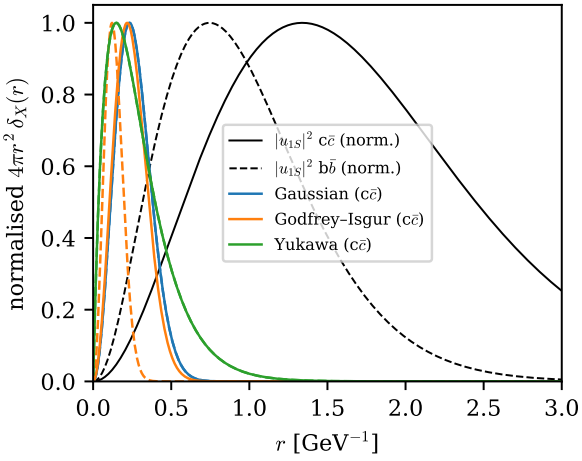


Figure 3: The fitted contact-term profiles $4\pi r^2 \delta_X(r)$ (fixed coupling, $\kappa = 0$) compared with the $1S$ wave functions. The fitted widths are far inside the wave functions.

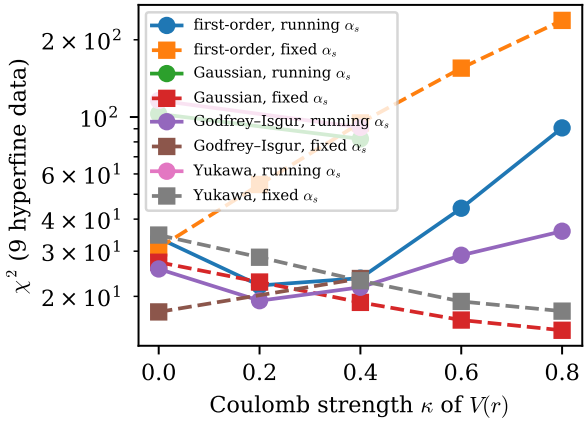


Figure 4: The χ^2 of Table 5 against κ. Circles and solid lines: running coupling; squares and dashed lines: fixed.

fixed coupling it selects $\kappa = 0$. Running and Coulomb strength are determined jointly, and only their one-gluon-exchange combination fits.

3. The $c\bar{c}$ $2S$ splitting is over-predicted by 12–15 MeV by every prescription, the size of the coupled-channel shift computed for the $\eta_c(2S)$ and $\psi(2S)$; it should not be used to tune contact interactions.

4. With it removed, first-order perturbation theory with a running coupling and no smearing describes the other eight observables with $\chi^2 = 2.8/7$ and one parameter. The simplest prescription is the right one.

5. It predicts $M(B_c^*) - M(B_c) = 62 \pm 1 \pm 3$ MeV and $M(B_c^*(2S)) - M(B_c(2S)) = 33 \pm 1 \pm 2$ MeV (64.5 and 34.4 once the η_c annihilation shift is taken out of the charmonium input), agreeing with ATLAS and CMS and 2σ above the lattice prediction.

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Code and data availability

The code, all fits and the scripts that regenerate every table and figure are available at `https://github.com/apoppong24/bc-hyperfine`; the spin-averaged potentials are those of Ref. [7].

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